

4.3 曲面的结构方程

任何光滑函数 f 的二阶偏导数可交换求导次序，即

$$\frac{\partial^2 f}{\partial u^\alpha \partial u^\beta} = \frac{\partial^2 f}{\partial u^\beta \partial u^\alpha}$$

而这对 $\mathbf{r}, \mathbf{r}_\alpha, \mathbf{n}$ 同样适用，因此我们有

$$\frac{\partial}{\partial u^\beta} \left(\frac{\partial \mathbf{r}}{\partial u^\alpha} \right) = \frac{\partial}{\partial u^\alpha} \left(\frac{\partial \mathbf{r}}{\partial u^\beta} \right) \quad (3.1)$$

$$\frac{\partial}{\partial u^\gamma} \left(\frac{\partial \mathbf{r}_\alpha}{\partial u^\beta} \right) = \frac{\partial}{\partial u^\beta} \left(\frac{\partial \mathbf{r}_\alpha}{\partial u^\gamma} \right) \quad (3.2)$$

$$\frac{\partial}{\partial u^\beta} \left(\frac{\partial \mathbf{n}}{\partial u^\alpha} \right) = \frac{\partial}{\partial u^\alpha} \left(\frac{\partial \mathbf{n}}{\partial u^\beta} \right) \quad (3.3)$$

$\mathbf{r}, \mathbf{r}_\alpha, \mathbf{n}$ 满足运动方程

$$\frac{\partial \mathbf{r}}{\partial u^\alpha} = \mathbf{r}_\alpha, \quad \alpha = 1, 2; \quad (M_1)$$

$$\frac{\partial \mathbf{r}_\alpha}{\partial u^\beta} = \Gamma_{\alpha\beta}^\gamma \mathbf{r}_\gamma + b_{\alpha\beta} \mathbf{n}, \quad \alpha, \beta = 1, 2; \quad (M_2)$$

$$\frac{\partial \mathbf{n}}{\partial u^\alpha} = -b_\alpha^\beta \mathbf{r}_\beta, \quad \beta = 1, 2; \quad (M_3)$$

将 (M_1) 代入 (3.1) 得： $\mathbf{r}_{\alpha\beta} = \mathbf{r}_{\beta\alpha}$

将 (M_2) 代入 (3.2) ，有

$$\begin{aligned} \frac{\partial}{\partial u^\gamma} \left(\frac{\partial \mathbf{r}_\alpha}{\partial u^\beta} \right) &= \frac{\partial}{\partial u^\gamma} (\Gamma_{\alpha\beta}^\varepsilon \mathbf{r}_\varepsilon + b_{\alpha\beta} \mathbf{n}) \\ &= \frac{\partial \Gamma_{\alpha\beta}^\varepsilon}{\partial u^\gamma} \mathbf{r}_\varepsilon + \Gamma_{\alpha\beta}^\varepsilon \mathbf{r}_{\varepsilon\gamma} + \frac{\partial b_{\alpha\beta}}{\partial u^\gamma} \mathbf{n} + b_{\alpha\beta} \mathbf{n}_\gamma \end{aligned} \quad (3.4)$$

再利用 (M_2) 和 (M_3) 就有

$$\frac{\partial}{\partial u^\gamma} \left(\frac{\partial \mathbf{r}_\alpha}{\partial u^\beta} \right) = \left(\frac{\partial \Gamma_{\alpha\beta}^\varepsilon}{\partial u^\gamma} + \Gamma_{\alpha\beta}^\eta \Gamma_{\eta\gamma}^\varepsilon - b_{\alpha\beta} b_\gamma^\varepsilon \right) \mathbf{r}_\varepsilon + \left(\Gamma_{\alpha\beta}^\varepsilon b_{\varepsilon\gamma} + \frac{\partial b_{\alpha\beta}}{\partial u^\gamma} \right) \mathbf{n} \quad (3.5)$$

同理，将运动方程代入 (3.2) 式有

$$\begin{aligned}\frac{\partial}{\partial u^\beta} \left(\frac{\partial \mathbf{r}_\alpha}{\partial u^\gamma} \right) &= \frac{\partial \Gamma_{\alpha\gamma}^\varepsilon}{\partial u^\beta} \mathbf{r}_\varepsilon + \Gamma_{\alpha\gamma}^\varepsilon (\Gamma_{\varepsilon\beta}^\eta \mathbf{r}_\eta + b_{\varepsilon\beta} \mathbf{n}) + \frac{\partial b_{\alpha\gamma}}{\partial u^\beta} \mathbf{n} + b_{\alpha\gamma} (-b_\beta^\xi \mathbf{r}_\xi) \\ &= \left(\frac{\partial \Gamma_{\alpha\gamma}^\varepsilon}{\partial u^\beta} + \Gamma_{\alpha\gamma}^\eta \Gamma_{\eta\beta}^\varepsilon - b_{\alpha\gamma} b_\beta^\varepsilon \right) \mathbf{r}_\varepsilon + \left(\Gamma_{\alpha\gamma}^\varepsilon b_{\varepsilon\beta} + \frac{\partial b_{\alpha\gamma}}{\partial u^\beta} \right) \mathbf{n}\end{aligned}\quad (3.6)$$

由(3.5)与(3.6)等式相同，有

$$\frac{\partial \Gamma_{\alpha\beta}^\varepsilon}{\partial u^\gamma} - \frac{\partial \Gamma_{\alpha\gamma}^\varepsilon}{\partial u^\beta} + \Gamma_{\alpha\beta}^\eta \Gamma_{\eta\gamma}^\varepsilon - \Gamma_{\alpha\gamma}^\eta \Gamma_{\eta\beta}^\varepsilon - b_{\alpha\beta} b_\gamma^\varepsilon + b_{\alpha\gamma} b_\beta^\varepsilon = 0, \quad (3.7)$$

$$\frac{\partial b_{\alpha\beta}}{\partial u^\gamma} - \frac{\partial b_{\alpha\gamma}}{\partial u^\beta} + \Gamma_{\alpha\beta}^\varepsilon b_{\varepsilon\gamma} - \Gamma_{\alpha\gamma}^\varepsilon b_{\varepsilon\beta} = 0 \quad (3.8)$$

把 (M_3) 代入(3.5)式有

$$\frac{\partial b_\beta^\varepsilon}{\partial u^\gamma} - \frac{\partial b_\gamma^\varepsilon}{\partial u^\beta} = -b_\beta^\eta \Gamma_{\eta\gamma}^\varepsilon + b_\gamma^\eta \Gamma_{\eta\beta}^\varepsilon \quad (3.9)$$

利用 $b_\alpha^\beta = b_{\alpha\gamma} g^{\gamma\beta}$ ，可证(3.8)与(3.9)相互等价。

方程(3.7)为曲面的Guass方程；方程(3.9)为曲面的Codazzi方程，合称为Guass-Codazzi方程或曲面的结构方程。

由于 $\alpha, \beta, \gamma, \varepsilon = 1, 2$ ；Guass-Codazzi方程形式上复杂。引入Riemann记号：

$$R_{\delta\alpha\beta\gamma} = g_{\delta\varepsilon} \left(\frac{\partial \Gamma_{\alpha\beta}^\varepsilon}{\partial u^\gamma} - \frac{\partial \Gamma_{\alpha\gamma}^\varepsilon}{\partial u^\beta} + \Gamma_{\alpha\beta}^\eta \Gamma_{\eta\gamma}^\varepsilon - \Gamma_{\alpha\gamma}^\eta \Gamma_{\eta\beta}^\varepsilon \right) \quad (3.10)$$

Guass方程(3.7)可改为

$$R_{\delta\alpha\beta\gamma} = g_{\delta\varepsilon} (b_{\alpha\beta} b_\gamma^\varepsilon - b_{\alpha\gamma} b_\beta^\varepsilon) = -(b_{\alpha\gamma} b_{\beta\delta} - b_{\alpha\beta} b_{\gamma\delta}) \quad (3.11)$$

利用第二类Christoffel符号 $\Gamma_{\alpha\beta}^\varepsilon = g_{\varepsilon\xi} \Gamma_{\alpha\beta\xi}$ ，由(3.10)可知

$$\begin{aligned}R_{\delta\alpha\beta\gamma} &= \frac{\partial \Gamma_{\delta\alpha\beta}}{\partial u^\gamma} - \frac{\partial \Gamma_{\delta\alpha\gamma}}{\partial u^\beta} - \frac{\partial g_{\delta\eta}}{\partial u^\gamma} \Gamma_{\alpha\beta}^\eta + \frac{\partial g_{\delta\eta}}{\partial u^\beta} \Gamma_{\alpha\gamma}^\eta + \Gamma_{\alpha\beta}^\eta \Gamma_{\delta\eta\gamma} - \Gamma_{\alpha\gamma}^\eta \Gamma_{\delta\eta\beta} \\ &= \frac{\partial \Gamma_{\delta\alpha\beta}}{\partial u^\gamma} - \frac{\partial \Gamma_{\delta\alpha\gamma}}{\partial u^\beta} + \Gamma_{\alpha\beta}^\eta \left(\Gamma_{\delta\eta\gamma} - \frac{\partial g_{\delta\eta}}{\partial u^\gamma} \right) - \Gamma_{\alpha\gamma}^\eta \left(\Gamma_{\delta\eta\beta} - \frac{\partial g_{\delta\eta}}{\partial u^\beta} \right)\end{aligned}\quad (3.12)$$

故

$$R_{\delta\alpha\beta\gamma} = \frac{1}{2} \left(\frac{\partial^2 g_{\delta\beta}}{\partial u^\gamma \partial u^\alpha} - \frac{\partial^2 g_{\alpha\beta}}{\partial u^\delta \partial u^\gamma} - \frac{\partial^2 g_{\delta\gamma}}{\partial u^\alpha \partial u^\beta} + \frac{\partial^2 g_{\alpha\gamma}}{\partial u^\beta \partial u^\delta} \right) - \Gamma_{\alpha\beta}^\eta \Gamma_{\delta\eta\gamma} + \Gamma_{\alpha\gamma}^\eta \Gamma_{\delta\eta\beta} \quad (3.13)$$

Riemann记号满足 $R_{\delta\alpha\beta\gamma} = R_{\beta\gamma\delta\alpha} = -R_{\alpha\delta\beta\gamma} = -R_{\delta\alpha\gamma\beta}$ ，而(3.11)的右边也满足同样的对称性质，故(3.11)中只有一个独立方程。

$$R_{1212} = -(b_{11}b_{22} - (b_{12})^2) \quad (3.14)$$

当式子(3.8)中 $\beta = \gamma$ 时 Codazzi方程只有两个独立方程

$$\begin{cases} \frac{\partial b_{11}}{\partial u^2} - \frac{\partial b_{12}}{\partial u^1} = b_{1\varepsilon}\Gamma_{12}^\varepsilon - b_{2\varepsilon}\Gamma_{11}^\varepsilon \\ \frac{\partial b_{21}}{\partial u^2} - \frac{\partial b_{22}}{\partial u^1} = b_{1\varepsilon}\Gamma_{22}^\varepsilon - b_{2\varepsilon}\Gamma_{21}^\varepsilon \end{cases} \quad (3.15)$$

若曲面的参数 (u, v) 为正交参考系，Guass方程为

$$-\frac{1}{\sqrt{EG}} \left\{ \left(\frac{(\sqrt{G})_u}{\sqrt{E}} \right)_v + \left(\frac{(\sqrt{E})_v}{\sqrt{G}} \right)_u \right\} = \frac{LN - M^2}{EG} \quad (3.16)$$

Codazzi方程可简化为

$$\begin{cases} \left(\frac{L}{\sqrt{E}} \right)_v - \left(\frac{M}{\sqrt{E}} \right)_u - N \frac{(\sqrt{E})_v}{G} - M \frac{(\sqrt{G})_u}{\sqrt{EG}} = 0, \\ \left(\frac{N}{\sqrt{G}} \right)_u - \left(\frac{M}{\sqrt{G}} \right)_v - L \frac{(\sqrt{G})_u}{E} - M \frac{(\sqrt{E})_v}{\sqrt{EG}} = 0, \end{cases} \quad (3.17)$$